

Solutions to Mock JEE Main - 18 | JEE-2022

PHYSICS

SECTION-1

1.(A) $E_x = \frac{-dV}{dx} = \frac{4-2}{(6-4)10^{-2}} = -100Vm^{-1}$

$E_y = -\frac{dV}{dy} = \frac{2-4}{(2-1)10^{-2}} = 200Vm^{-1}$

2.(C) Internal resistance of cell can be calculated from the relation

$0.3 = \frac{1.5}{4+r} \text{ i.e. } r = 1\Omega$

Now when 4Ω resistor is also introduced total resistance in the external circuit is 3Ω

$\left[\because \frac{4 \times 4}{4+4} + 1 = 3\Omega \right]$

$\therefore$ Circuit current $= \frac{1.5}{3} = 0.5A$ and current through meter $= \frac{0.5}{2} = 0.25A$

When voltmeter is connected. $I = \frac{1.5}{4+1} = 0.3A$

Then $V = E - Ir = 1.5 - 0.3 \times 1 = 1.2V$

3.(A) At steady state, although no current passes through the capacitor yet it creates a p.d. between a and b

Here $i_1 = 2 + 1 = 3A$

and $i_2 + 1 = 2 \text{ i.e., } i_2 = 1A$

In mesh abefa $i_1 \times 5 + i_1 \times 1 + i_2 \times 2 = V_a - V_b$ or $3 \times 5 + 3 \times 1 + 2 = V_a - V_b$

i.e., $V_a - V_b = 20V$

Energy stored, $E = \frac{1}{2}CV^2 = \frac{1}{2} \times 4 \times 10^{-6} \times 20^2 = 8 \times 10^{-4} J$

4.(B) True value of angle of dip, $\phi = 60^\circ$

Angle of inclination of plane, $\alpha = 30^\circ$

Let apparent dip be δ , then $\tan \delta = \frac{\tan \phi}{\cos \alpha} = \frac{\tan 60^\circ}{\cos 30^\circ}$

$= \frac{\sqrt{3}}{2} = 2$; $\delta = \tan^{-1} 2$.

5.(A) $\vec{\tau} = \vec{M} \times \vec{B}$ or $\tau = MB \sin \theta$

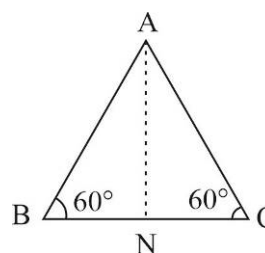
$M = \text{Dipole moment of triangle} = IA = I \times \frac{1}{2}(AN)(BC)$

$= I \times \frac{1}{2}(AN)(2BN) = I \times (AN)(BN)$

$= I(AB \sin 60^\circ)(AB \cos 60^\circ) = 0.1 \times \left(0.02 \times \frac{\sqrt{3}}{2}\right) \times \left(0.02 \times \frac{1}{2}\right)$

$= 1.73 \times 10^{-5} Am^2$; $B = 5 \times 10^{-2} T$ and $\theta = 90^\circ$

$\therefore \tau = 1.73 \times 10^{-5} \times 5 \times 10^{-2} \times \sin 90^\circ = 8.65 \times 10^{-7} Nm$



6.(C) Using the relation $E = h\nu$, $\nu = \frac{E}{h}$ $\left(E = 4.4 \text{ keV} = 44 \times 10^3 \times 1.6 \times 10^{-19} \text{ J} \right)$

$$= \frac{4.4 \times 10^3 \times (1.6 \times 10^{-19})}{6.6 \times 10^{-34}} \simeq 1 \times 10^{12} \text{ MHz}$$

Part of the spectrum for this frequency is 'X-ray'

7.(D) When plane surface is polished, $\frac{1}{F'} = \frac{1}{30} = \frac{2(\mu - 1)}{R}$

$$\frac{1}{F'} = \frac{1}{10} = \frac{2\mu}{R}$$

Solving equations (i) and (ii) $\mu = 1.5$

8.(B) Resolution limit of eye $= \left(\frac{1}{60} \right)^0 = \frac{1}{60} \times \frac{\pi}{180}$

Let minimum distance between poles be d then $\frac{d}{11000} = \frac{1}{60} \times \frac{\pi}{180}$

or $d = (11000) \times \frac{1}{60} \times \frac{\pi}{180} = 3 \text{ m}$.

9.(A) Initial fringe width $\beta = \frac{\lambda D}{d} = \frac{\lambda D}{10^{-3}}$... (i)

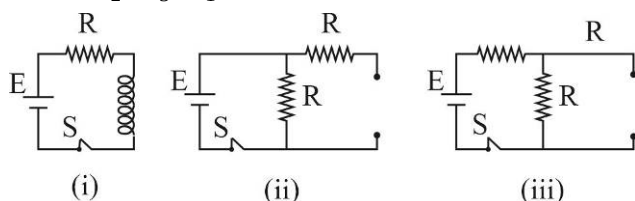
Final fringe width, $\left(\beta - 3 \times 10^{-5} \right) = \frac{\lambda (D - 5 \times 10^{-2})}{10^{-3}}$... (ii)

Solving (i) and (ii) $\lambda = 6 \times 10^{-7} \text{ m}$

10.(B) At $t = 0$, current through L is zero, so it acts as open circuit. The given figures can be redrawn as shown.

$$i_1 = 0, i_2 = \frac{E}{R}, i_3 = \frac{E}{2R}$$

Hence, $i_2 > i_3 > i_1$.



11.(B) Initially, the current lags behind the potential difference. Hence, the circuit contains resistance and inductance. The power of the circuit is

$$P = V_{rms} \times i_{rms} \times \cos \phi$$

But, $i_{rms} = \frac{V_{rms}}{Z}$ or $Z = \frac{(V_{rms})^2 \times \cos \phi}{P} = \frac{(220)^2 \times 0.8}{550} = 70.4 \Omega$

power factor, $\cos \phi = \frac{R}{Z} \therefore R = Z \cos \phi = 70.4 \times 0.8 = 56.32 \Omega$

Now, $Z^2 = R^2 + (\omega L)^2$

$\therefore (\omega L)^2 = Z^2 - R^2 = (70.4)^2 - (56.4)^2 = 1775.2 \therefore \omega L = 42.2 \Omega$

The impedance of the circuit after inserting the capacitance is given by

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

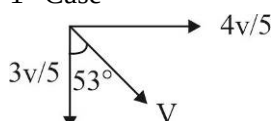
Clearly, for making power factor = 1, it must be that $\omega L = \frac{1}{\omega C}$ or $C = \frac{1}{\omega(\omega L)}$

But, $\omega = 2\pi f = 2 \times 3.14 \times 50 = 314$ $\therefore C = \frac{1}{314 \times 42.2} = 75 \times 10^{-6} F = 75 \mu F$

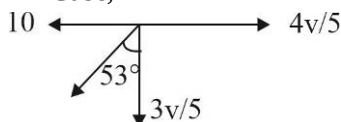
12.(C) $\delta_{net} = \delta_{mirror} + \delta_{prism} = (180 - 2i) + (\mu - 1)A$
 $(180^\circ - 2 \times 45^\circ) + (1.5 - 1) \times 4^\circ = 92^\circ$

13.(C) 14.(C)

1st Case



2nd Case,



$$10 - \frac{4v}{5} = \frac{4v}{5}$$

$$\frac{8v}{5} = 10; \quad v = \frac{25}{4}$$

15.(A) For a small strip at a distance x from vertex, area of strip is $dA = \frac{b}{h} x dx$

$$\text{Field of wire at location of the strip } B = \frac{\mu_0 I}{2\pi(a+x)}$$

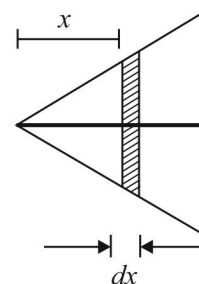
Flux linked with strip is

$$d\phi = BdA = \frac{\mu_0 I}{2\pi(a+x)} \cdot \frac{b}{h} x dx = \frac{\mu_0 Ib}{2\pi h} \left(\frac{x dx}{a+x} \right)$$

$$\text{Total flux linked with triangular coil, } \phi = \frac{\mu_0 Ib}{2\pi h} \int_0^h \frac{x dx}{a+x} = \frac{\mu_0 Ib}{2\pi h} \left[h - a \ln \left(\frac{a+h}{a} \right) \right]$$

$$\therefore \text{Mutual inductance of the coil is } M = \frac{\phi}{I} = \frac{\mu_0 b}{2\pi h} \left(h - a \ln \left(\frac{a+h}{a} \right) \right)$$

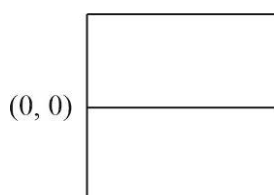
$$\Rightarrow M = \frac{4\pi \times 10^{-7} \times 0.2}{2\pi \times 0.1} \left(0.1 - 0.1 \ln \left(\frac{10+10}{10} \right) \right) = 1.22 \times 10^{-8} H$$



16.(B) $x_{cm} = \frac{mx_0 + 2m \times l/2}{3m} = \frac{l}{3}$

$$y_{cm} = \frac{2mx_0 + m + l/2}{3m} = \frac{l}{6}$$

$$COM = \left(\frac{l}{3}, \frac{l}{6} \right)$$



17.(A) Since 10 line are emitted hence e^- is in $n=5$ to $n=4$

$$\frac{1}{\lambda} = R \left[\frac{1}{16} - \frac{1}{25} \right]; \quad \lambda = 4.05 \times 10^{-6} m$$

18.(A) Let the second stone crosses first stone at a distance x from the top, then

$$\text{For first stone } x = \frac{1}{2} g (t+n)^2 \quad \dots(i)$$

$$\text{and for second stone } x = ut + \frac{1}{2} gt^2 \quad \dots(ii)$$

$$\text{From Equations. (i) and (ii) } \frac{1}{2} g (t+n)^2 = ut + \frac{1}{2} gt^2$$

$$\Rightarrow t = \frac{gn^2}{2(u-gn)}; \quad x = \frac{g}{2} \left[\frac{n \left(\frac{gn}{2} - u \right)}{gn - u} \right]^2$$

19.(B) $g \propto \rho R \Rightarrow \rho_1 = \rho_2 = \rho_3$

$$g_1 : g_2 : g_3 = R_1 : R_2 : R_3 \Rightarrow g_1 : g_2 : g_3 = 6 : 3 : 2$$

$$\text{and } v = \sqrt{2gR} \Rightarrow v \propto \sqrt{gR} \Rightarrow v \propto R \quad \therefore v_1 : v_2 : v_3 = 6 : 3 : 2$$

20.(C) According to the problem, power = 1 Watt

Zener breakdown voltage $V_z = 5V$

Minimum voltage $V_{\min} = 3V$

Maximum voltage $V_{\max} = 7V$

$$\text{We know, } P = VI; \quad \text{So current } I_{Z_{\max}} = \frac{P}{V_z} = \frac{1}{5} = 0.2A$$

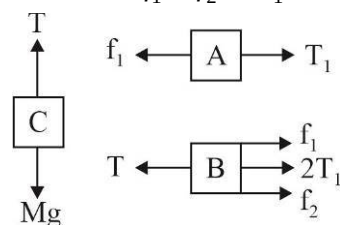
$$\text{For safe operation, } R_s \text{ will be equal to } R_s = \frac{V_{\max} - V_z}{I_{Z_{\max}}} = \frac{7-5}{0.2} = \frac{2}{0.2} = 10\Omega$$

SECTION-2

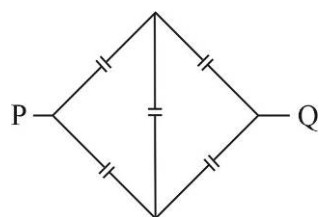
21.(9) for C, $T = Mg$

for A, $T_1 = f_1$

for B, $T = f_1 + f_2 + 2T_1$



22.(5) Equivalent circuit $C_{eq} = 25pF$



$$23.(5) \quad R_{\text{heating filament}} = \frac{V^2}{P} = 10\Omega; \quad R_{\text{parallel}} = 5\Omega$$

24.(3) When $m_2 = 3m_1$, m_2 comes to rest after collision with m_1 .
 $\therefore$ KE transferred = 100%

$$25.(4) \quad R_{\text{approx}} = \frac{120}{2} = 60\Omega$$

$$i_X = i - i_V = 2 - \frac{120}{960} = \frac{15}{8} A$$

$$\Rightarrow R = \frac{120}{\left(\frac{15}{8}\right)} = 64\Omega \quad \Rightarrow \Delta R = 4\Omega$$

26.(1) Initially the rod will be in equilibrium if $2T_0 = Mg$ with $T_0 = kx_0$... (i)

When the current I is passed through the rod, it will experience a force $F = BIL$ vertically up,

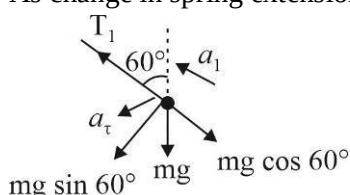
In equilibrium $2T + BIL = Mg$ with $T = kx$... (ii)

From (i) and (ii)

$$\frac{T}{T_0} = \frac{Mg - BIL}{Mg} \text{ i.e. } \frac{x}{x_0} = 1 - \frac{BIL}{Mg} \quad \text{or, } B = \frac{Mg(x_0 - x)}{IL x_0}$$

Putting the value we get $B = 0.5 \times 10^{-2} T$.

27.(3) As change in spring extension is zero FBD of A



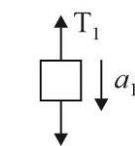
$$T_1 - mg \cos 60^\circ = ma_1$$

$$a_\tau = \frac{mg \sin 60^\circ}{m} = g \frac{\sqrt{3}}{2}$$

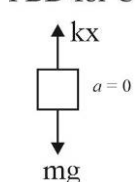
$$mg + kx - T_1 = ma_1$$

$$a_1 = \frac{3g}{4}$$

FBD of B



FBD for C



$$28.(4) \quad T = 2ma_c$$

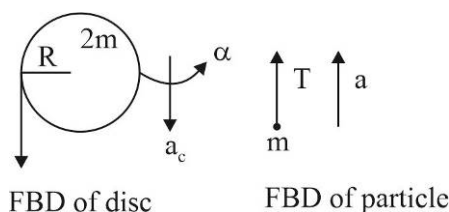
$$RT = \frac{2mR^2}{2} \alpha$$

$$T + m(a_c + \alpha R) = \frac{mJ^2}{m^2 2R}$$

$$T + \frac{T}{2} + T = \frac{J^2}{2mR}$$

$$\therefore \frac{5T}{2} = \frac{J^2}{2mR} \Rightarrow T = \frac{J^2}{5mR}$$

$$a_c = \frac{T}{2m} = \frac{J^2}{5mR2m} = \frac{J^2}{10m^2 R} \text{ putting all the values } a_c = 4m/s^2.$$

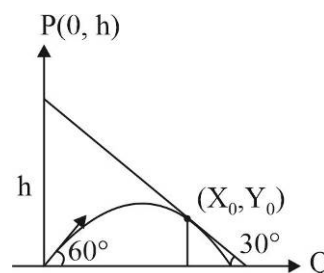


$$29.(2) \quad y = \sqrt{3}x - \frac{x^2}{5} \Rightarrow \frac{dy}{dx} = \sqrt{3} - \frac{2x}{5}; \tan(-30^\circ) = -\frac{1}{\sqrt{3}}$$

$$\frac{2x}{5} = \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}} \Rightarrow x = \frac{10}{\sqrt{3}} m \Rightarrow y = 10 - \frac{100}{5 \times 3} = \frac{10}{3} m$$

$$\text{Equation of PQ } y - \frac{10}{3} = \frac{-1}{\sqrt{3}} \left(x - \frac{10}{\sqrt{3}} \right)$$

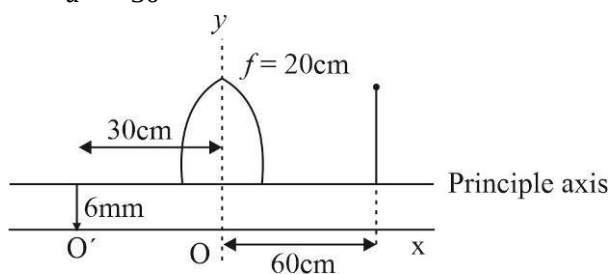
$$\text{For point } P: x = 0, y = h \Rightarrow h - \frac{10}{3} = \frac{10}{3} \Rightarrow h = \frac{20}{3} m$$



$$30.(2) \Rightarrow \frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{30} + \frac{1}{20} = \frac{3-2}{60} = \frac{1}{60}$$

$$\Rightarrow v = 60 \text{ cm}$$

$$m = \frac{v}{u} = \frac{60}{-30} = -2$$



$$\text{Height of image} = (-2)(-6) = 12 \text{ mm}$$

$$\text{Y-coordinate of the image of } Q_1 = 12 + 6 = 18 \text{ mm} = 9 \times 2 \text{ mm}$$

Chemistry

SECTION-1

1.(B) No. of radial nodes = $n - l - 1$

2.(A) $2\text{H}_2(\text{g}) \longrightarrow 4\text{H}^+ + 4\text{e}^- \quad E^\circ = 0.0\text{V}$

$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O}(\text{l}) \quad E^\circ = 1.23\text{V}$

$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O}(\text{l}) \quad E^\circ_{\text{cell}} = 1.23\text{V}$

$\Delta G^\circ_{298} = -nFE^\circ = -4 \times 96500 \times 1.23 = -474.78\text{kJ}$

$\Delta G^\circ_{298} = \Delta H^\circ_{298} - T\Delta S^\circ_{298}$

$\Delta S^\circ_{298} = \frac{2 \times (-285.5) - (-475)}{298} = -0.322\text{kJ/K}$

3.(C) $\text{Ag}^+ + 2\text{NH}_3 \longrightarrow \text{Ag}(\text{NH}_3)_2^+$

$0.8 - x \quad z - 2x \quad x; \quad K_f = 10^8; \text{ so } x \approx 0.8$

$5 \times 10^{-8} z - 0.8 \times 2 \quad 0.8$

$K_f = \frac{[\text{Ag}(\text{NH}_3)_2^+]}{[\text{Ag}^+][\text{NH}_3]^2} = \frac{0.8}{5 \times 10^{-8} (z - 1.6)^2} = 10^8$

$(z - 1.6)^2 = 0.16; z - 1.6 = 0.4; z = 2 \quad \therefore V = 6\text{ litre}, n = 12$

4.(A) $3\text{A}_{(\text{g})} \longrightarrow 2\text{B}_{(\text{g})} + 2\text{C}_{(\text{g})}$

$t = 0 \quad P_0 \quad 1 \quad 1$

$t = t \quad P_0 - X \quad \frac{2}{3}X \quad \frac{2}{3}X$

After $0 \quad \frac{2}{3}P_0 \quad \frac{2}{3}P_0$

Long time

$\frac{4}{3}P_0 = 4 \Rightarrow P_0 = 3; \quad P_0 + \left(\frac{4}{3} - 1\right)X = P_0 + \frac{1}{3}X = 3.5; \quad \frac{1}{3}X = 0.5 \Rightarrow X = 1.5$

In 20 min. pressure of A decreased from 3 to 1.5 atm so $t_{1/2} = 20\text{ min}$ time taken for 25% decomposition of A is equal to 10 minute.

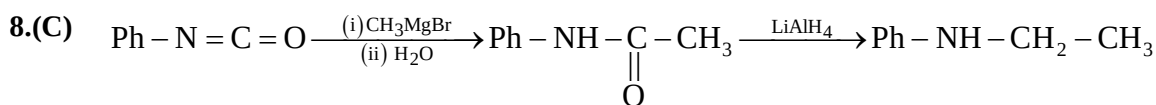
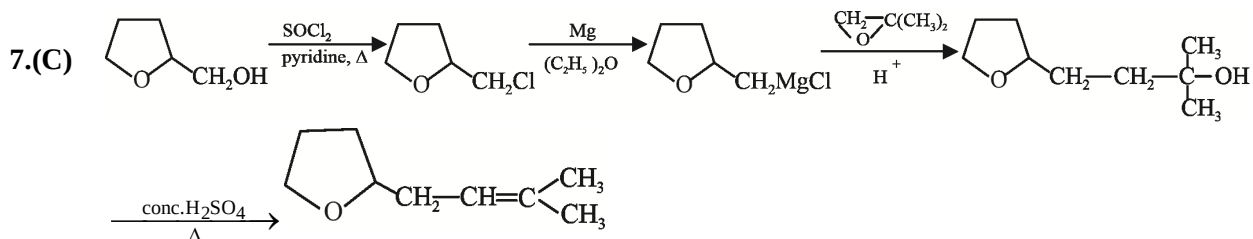
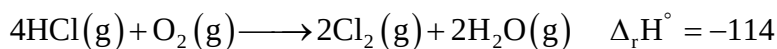
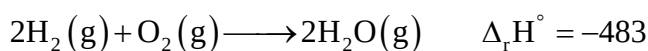
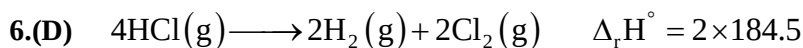
5.(B) $\text{A}_2\text{B}_3(\text{aq}) \longrightarrow 2\text{A}^{3+}(\text{aq}) + 3\text{B}^{2-}(\text{aq})$

$n = 5$

so, $i = 1 + (5 - 1) \times \alpha$

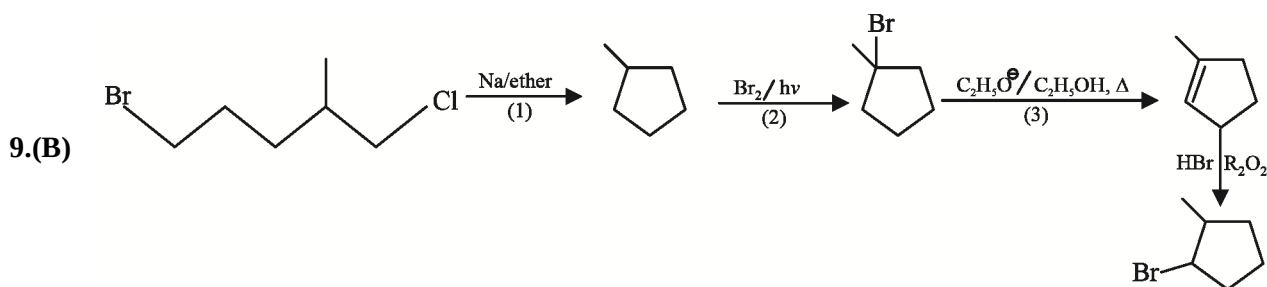
$\Delta T_b = iK_b m = (1 + 4\alpha)K_b m$

$= (1 + 4 \times 0.7) \times 0.5 \times 1 = 1.9; \quad T_b = 373 + 1.9$



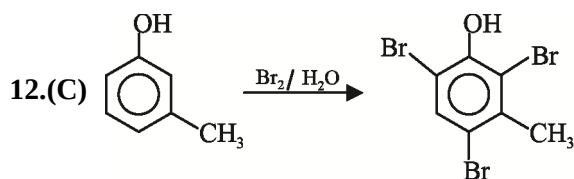
“P” (2° Amide)

“Q” (2° Amine)



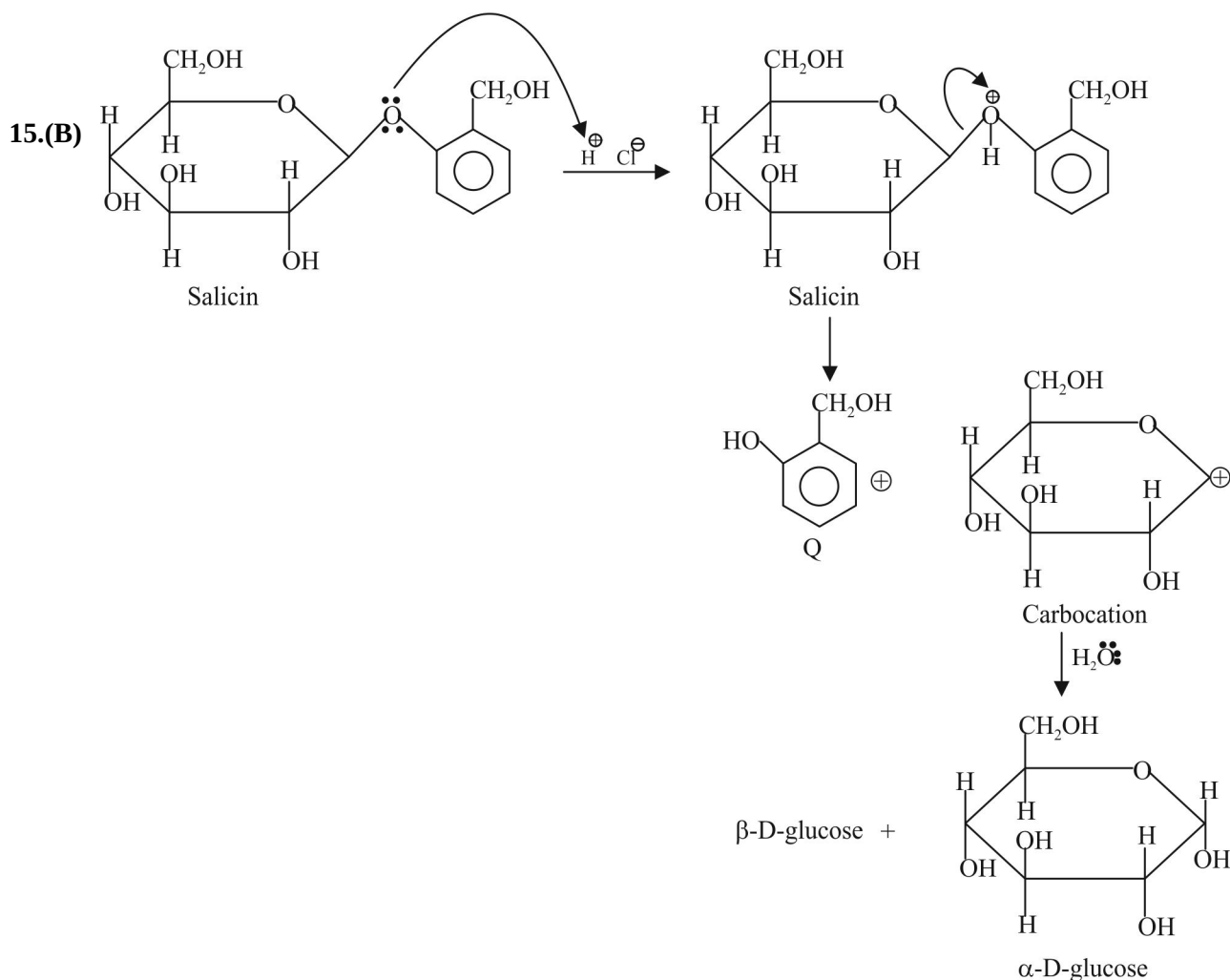
10.(B) –M group destabilizes the carbocation therefore slowest rate of electrophilic addition.

11.(A) $[\text{MnO}_4]^{2-}$; Mn is in +6 oxidation state. Electronic configuration is $[\text{Ar}]^{18} 3d^1 4s^0$. As it contains one unpaired electron it is paramagnetic and green coloured because of d-d transition in visible region. $[\text{TiCl}_6]^{2-}$, $[\text{VO}_4]^{3-}$ and CrO_2Cl_2 are diamagnetic because all electrons are paired.

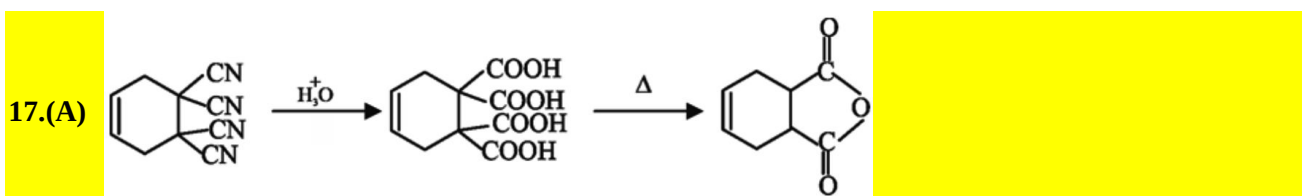


13.(B) Perkin's reaction + Markownikov's addition of HBr

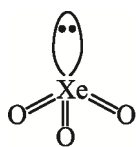
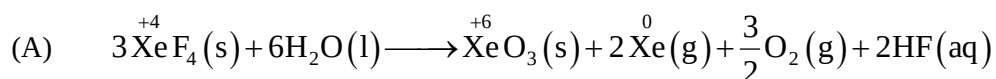
14.(A) Statement III, IV are incorrect.



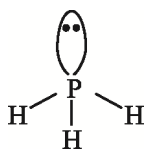
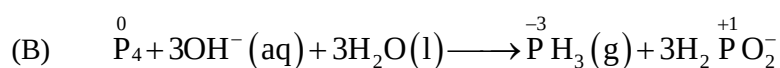
16.(A) FeSiO_3 is obtained as slag in the extraction of copper from copper pyrites.



18. (B) (A $\rightarrow$ p, r, s; B $\rightarrow$ p, r, s, t; C $\rightarrow$ q, r; D $\rightarrow$ p, r, s)

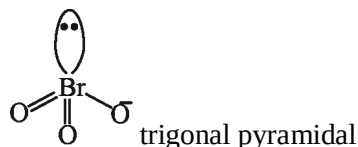
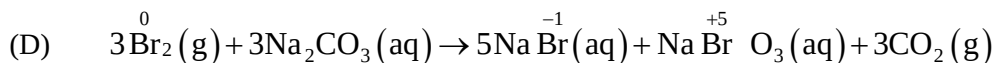
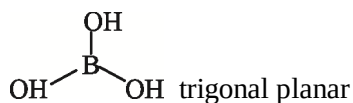
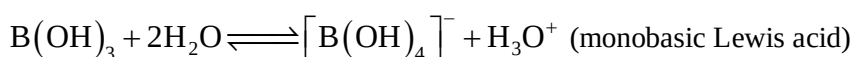
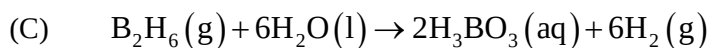


trigonal pyramidal



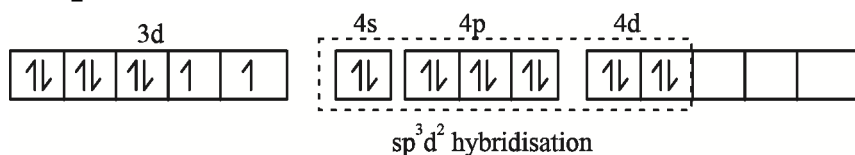
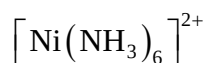
trigonal pyramidal

The spontaneous combustion of phosphine evolved dense white fumes which serves as signals.



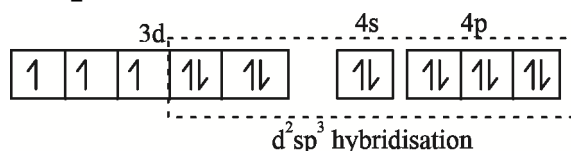
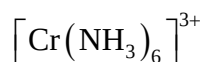
19.(C) Statement III is incorrect as with KI, Cu^{2+} salts form a white precipitate of Cu_2I_2 and I_2 gas is liberated.

20.(B) $_{28}Ni$ is in +2 oxidation state with electron configuration $[Ar]^{18} 3d^8 4s^0$, so



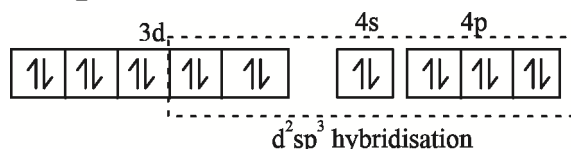
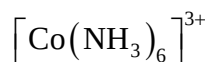
Number of unpaired electrons = 2

(b) $_{24}Cr$ is in +3 oxidation state with electron configuration $[Ar]^{18} 3d^3 4s^0$; so



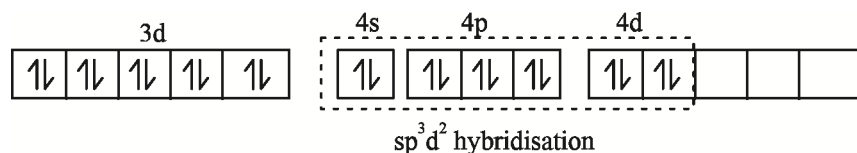
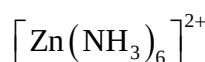
Number of unpaired electrons remains three, whether ligand is strong field or weak ligand when electron configuration is $3d^3$.

(C) $_{27}Co$ is in +3 oxidation state with electron configuration $[Ar]^{18} 3d^6 4s^0$, so



All electrons are paired, so complex ion is diamagnetic.

(d) $_{30}Zn$ is in +2 oxidation state with electron configuration $[Ar]^{18} 3d^{10} 4s^0$, so



SECTION-2

- 21.(12) In truncated octahedron: Hexagonal faces = 8
 In truncated tetrahedron: Hexagonal faces = 4
 Total hexagonal faces = 12

22.(100) $PV = nRT$; $1 \times V = \frac{1}{10} \times R \times (t + 273)$

There pressure now is 1.1 atm. $\therefore 1.1 \times V = \frac{1}{10} \times R (t + 283)$.

Solving $T = 100$

23.(10) Milli mole of NaCl = $M \times V = 0.5 \times 1 = 0.5$; Coagulation value = $\frac{0.5 \times 1000}{50} = 10$

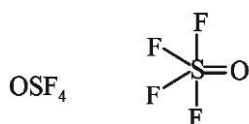
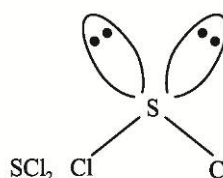
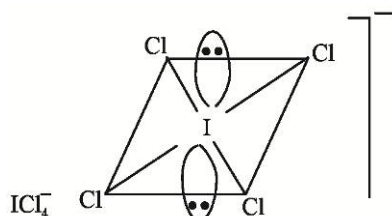
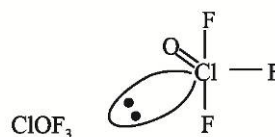
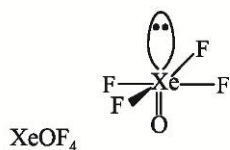
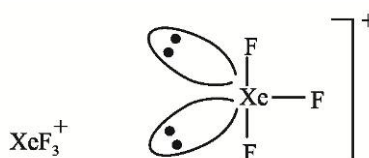
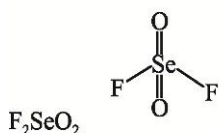
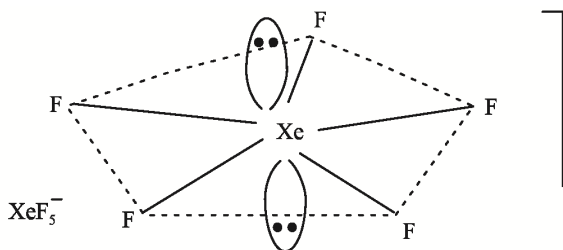
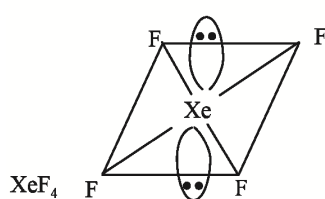
24.(14) Species Number of π^* electrons

O_2 2

O_2^- 3

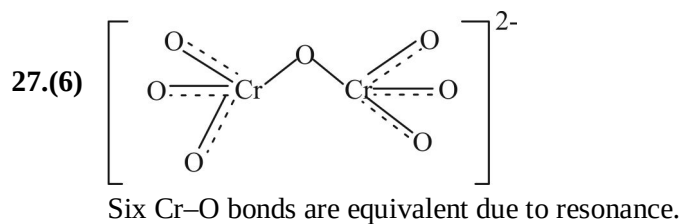
O_2^{2-} 4

Total = 9 electrons



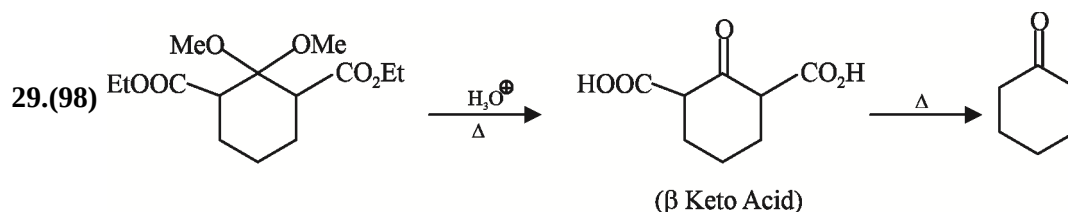
- 25.(4) (i) Argentite = Ag_2S , (ii) Calamine = ZnCO_3 (iii) Malachite = $\text{CuCO}_3, \text{Cu}(\text{OH})_2$
 (iv) Siderite = FeCO_3 (v) Galena = PbS , (iv) Magnetite = Fe_3O_4 ,
 (vii) Dolomite = $\text{MgCO}_3, \text{CaCO}_3$ (viii) Cassiterite = SnO_2 , (ix) Bauxite = $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$

- 26.(8) Phosphorous acid (H_3PO_3) Dibasic $\therefore x = 2$
 Tetrathionic acid ($\text{H}_2\text{S}_4\text{O}_6$) Dibasic $\therefore y = 2$
 Pyrophosphoric acid ($\text{H}_4\text{P}_2\text{O}_7$) tetrabasic $\therefore z = 4$



- 28.(8) Product = $\text{Ph} - \text{CH} = \text{CH} - \underset{\text{CH}_3}{\text{CH}} - \text{CH} = \text{NOH}$

Total number of stereoisomers = $2^n = 2^3 = 8$
 (n = number of stereo centres (G.I. + O.I.) = 2 + 1 = 3)



30.(100) $\text{X}_{(s)} \rightleftharpoons \text{A}_{(g)} + \text{B}_{(g)} \quad k_{p1} = 500 \text{ atm}$

$$P_1 + P_2 \quad P_1$$

$\text{Y}_{(s)} \rightleftharpoons \text{A}_{(g)} + \text{C}_{(g)} \quad k_{p2} = 2000 \text{ atm}$

$$P_1 + P_2 \quad P_2$$

$$(P_1 + P_2)P_1 = 500$$

$$(P_1 + P_2)P_2 = 2000$$

$$P_2 / P_1 = 4 \Rightarrow P_2 = 4P_1$$

$$5P_1^2 = 500$$

$$P_1^2 = 100 \Rightarrow P_1 = 10$$

$$P_2 = 40$$

$$\text{Total pressure} = 2(P_1 + P_2) = 2 \times 50 = 100$$

Mathematics

SECTION-1

- 1.(C) Let $z = \cos \theta + i \sin \theta$
 $\Rightarrow \sin \theta, \cos \theta \in Q$ (rational number)

Now $|z^{2n} - 1|$
 $= |\cos 2n\theta - 1 + i \sin 2n\theta|$
 $= |-2\sin^2 n\theta + 2i \sin n\theta \cos n\theta|$
 $= |2\sin n\theta| |-\sin n\theta + i \cos n\theta|$
 $= |2\sin n\theta| \in Q$
 $\Rightarrow x \in Q$

Hence, $(-1)^x$ is complex number.

- 2.(A) $\frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \frac{9}{400} + \frac{11}{900} + \dots \infty$

General term, $T_r = \frac{2r+1}{[r(r+1)]^2}$
 $= \frac{1}{r^2} - \frac{1}{(r+1)^2}$

$\therefore S_n = 1 - \frac{1}{(n+1)^2} \quad \therefore \lim_{n \rightarrow \infty} S_n = 1$

- 3.(A) For filling 16 gaps with 25 dissimilar things when repetition is allowed, number of ways = 25^{16}
 For filling 17 gaps with 25 dissimilar things when repetition is allowed, number of ways = 25^{17}

.....

Thus, required number of ways = $25^{16} + 25^{17} + \dots + 25^{25}$

$$= \frac{25^{16}(25^{10} - 1)}{25 - 1}$$

$$= \frac{25}{24}(25^{25} - 25^{15})$$

- 4.(A) $\log_{1/10}(x^2 + x) > \log_{1/2}(x^3 - x) + \log_2(x - 1)$

$$\Rightarrow \log_{1/10}(x^2 + x) > \log_2\left(\frac{x-1}{x(x^2-1)}\right) \Rightarrow \log_{1/10}(x^2 + x) > \log_2\left(\frac{1}{x^2 + x}\right)$$

$$\Rightarrow \log_{1/10}(x^2 + x) > \log_{1/2}(x^2 + x) \Rightarrow x^2 + x > 1$$

$$\Rightarrow x > \frac{-1 + \sqrt{5}}{2} \text{ or } x < \frac{-1 - \sqrt{5}}{2}$$

$$\text{But } x - 1 > 0 \Rightarrow x \in (1, \infty)$$

5.(D) We have

$$\begin{aligned} |z_1|^2 + |z_2|^2 + |z_3|^2 - |z_1||z_2| - |z_2||z_3| - |z_3||z_1| &= 0 \\ \Rightarrow \frac{1}{2} (|z_1| - |z_2|)^2 + (|z_2| - |z_3|)^2 + (|z_3| - |z_1|)^2 &= 0 \\ \Rightarrow |z_1| - |z_2| = 0, |z_2| - |z_3| = 0 \text{ and } |z_3| - |z_1| &= 0 \\ \Rightarrow |z_1| = |z_2| = |z_3| \end{aligned}$$

Thus, ABC is an equilateral triangle.

6.(A) $(1-x)^{200} = {}^{200}C_0 - {}^{200}C_1x + {}^{200}C_2x^2 - {}^{200}C_3x^3 + \dots \dots \dots$... (i)

and $\left(\frac{1-x^{149}}{1-x}\right) = (x^{148} + x^{147} + x^{146} + \dots + x + 1)$... (ii)

On multiplying and equating coefficient of x^{148} , we get

$$\begin{aligned} & {}^{200}C_0 - {}^{200}C_1 + {}^{200}C_2 \dots + {}^{200}C_{148} \\ &= \text{Coefficient of } x^{148} \text{ in } \left[(1-x)^{200} \cdot \frac{(1-x^{149})}{(1-x)} \right] \\ &= \text{Coefficient of } x^{148} \text{ in } (1-x)^{199} (1-x^{149}) \\ &= \text{Coefficient of } x^{148} \text{ in } (1-x)^{199} = {}^{199}C_{148} \end{aligned}$$

7.(D) S, S, E, E, R, I

Case I : when all letters are distinct

$${}^4C_3 \times 3! = 24$$

Case II : when 2 letters are same.

$${}^2C_1 \times {}^3C_1 \times \frac{3!}{2!} = 2 \times 3 \times 3 = 18$$

Total number of numbers = $24 + 18 = 42$

8.(C) $\alpha + \beta = 7$

$$\begin{aligned} \Rightarrow \frac{(\alpha^{10} - \beta^{10}) - (\alpha^8 - \beta^8)}{\alpha^{(\alpha+\beta+2)} - \beta^{(\alpha+\beta+2)}} \\ = \frac{\alpha^8(\alpha^2 - 1) - \beta^8(\beta^2 - 1)}{\alpha^9 - \beta^9} = \frac{7\alpha^9 - 7\beta^9}{\alpha^9 - \beta^9} = 7 \end{aligned}$$

9.(A) $\lim_{x \rightarrow 0^+} \left(\min. (t^2 - 6(t+x) + 12) + \left\lceil \frac{\log_e(1+2x)}{x} \right\rceil \right)$

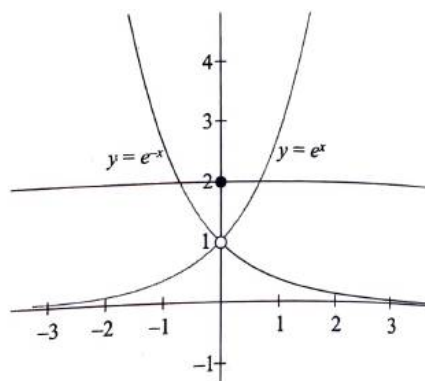
$$\begin{aligned} &= \lim_{x \rightarrow 0^+} \left(\min. (t^2 - 6(t+x) + 12) + \left\lceil \frac{\log_e(1+2x)}{2x} \cdot 2 \right\rceil \right) \\ &= \min. (t^2 - 6t + 12) + [2^-] \left(\because \lim_{x \rightarrow 0^+} \frac{\log_e(1+2x)}{2x} = 1 (< 1) \right) \\ &= \min. ((t-3)^2 + 3) + 1 = 4 \end{aligned}$$

10.(C) From the graph, $f(x) = \begin{cases} e^{-x}, & x \leq -\log_e 2 \\ 2 & -\log_e 2 < x \leq 0 \\ e^{-x}, & x > 0 \end{cases}$

Clearly $f(x)$ is discontinuous at $x = 0$

$f(x)$ is continuous but non-derivable at $x = \log_e \frac{1}{2}$

$f(x)$ has non-removable type of discontinuity at $x = 0$ with jump of discontinuity equal to 1.



11.(A) $f(x) = \sin \frac{\{x\}}{a} + \cos \frac{\{x\}}{a}, a > 0$

It attains max. if $\frac{\{x\}}{a} = \frac{\pi}{4}$

Now $\frac{\{x\}}{a} \in \left[0, \frac{1}{a}\right)$

$\Rightarrow \frac{1}{a} > \frac{\pi}{4}, \text{ for } f \text{ to have its maxima} \Rightarrow 0 < a < \frac{4}{\pi}$

12.(B) $\frac{1}{f(x)\{1+f^2(x)\}} = \frac{1}{\sqrt{x}(1+x)} \Rightarrow f(x) = \sqrt{x} \quad \therefore I = \int \frac{dx}{\sqrt{x}+x} = 2 \log |1+\sqrt{x}| + c$

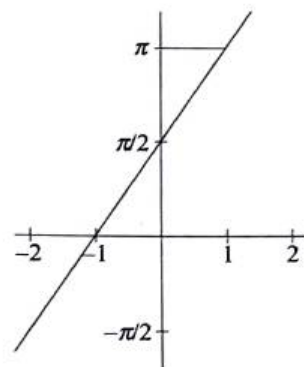
13.(B) $f(x) = (\sin^{-1} x + \cos^{-1} x) + (\tan^{-1} x + \tan^{-1}(1/x))$

$= \begin{cases} \frac{\pi}{2} + \frac{\pi}{2}, & x > 0 \\ \frac{\pi}{2} - \frac{\pi}{2}, & x < 0 \end{cases}$

$= \begin{cases} \pi, & x > 0 \\ 0, & x < 0 \end{cases}$

and $y = \frac{\pi}{2}x + \frac{\pi}{2}$

Graph of the function is as shown in the following figure.



From the figure, the required area is $2 \left(\frac{\pi}{2} \times 1 \times \frac{1}{2} \right) = \frac{\pi}{2}$

$$14.(A) \quad L_1: \frac{x-1}{2} = \frac{y-1}{1} = \frac{z-2}{0}$$

$$\text{and } L_2: \frac{x-2}{1} = \frac{y-3}{2} = \frac{z-2}{3}$$

$$\Rightarrow \quad \text{Normal to plane} = (2\hat{i} + \hat{j}) \times (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{vmatrix}$$

$$= 3(\hat{i} - 2\hat{j} + \hat{k})$$

$$\therefore \quad \text{DCs of normal vector } \frac{3(\hat{i} - 2\hat{j} + \hat{k})}{3\sqrt{6}} = \left\{ \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\}$$

$$15.(C) \quad \text{We have } y - x \frac{dy}{dx} = ay^2 + \frac{a}{dx}$$

$$\Rightarrow \quad (a+x) \frac{dy}{dx} = y(1-ay) \quad \Rightarrow \quad \frac{dy}{y(1-ay)} = \frac{dx}{a+x}$$

$$\Rightarrow \quad \int \left(\frac{1}{y} + \frac{a}{1-ay} \right) dy = \int \frac{dx}{a+x} \quad \Rightarrow \quad \log y - \log(1-ay) = \log(a+x) + \log c$$

$$\Rightarrow \quad y = c(1-ay)(x+a)$$

$$16.(A) \quad E_i = \text{Bag has exactly } i \text{ black balls}$$

$$3 \leq i \leq 10; \quad P(E_i) = \frac{1}{8}$$

$$E = \text{Three balls drawn are black}$$

$$P(E_9 / E) = \frac{P(E_9)P(E / E_9)}{\sum_{i=3}^9 P(E_i)P(E / E_i)}$$

$$\begin{aligned} &= \frac{\frac{1}{8} \frac{{}^9C_3}{{}^{10}C_3}}{\frac{1}{8} \left[\frac{{}^3C_3}{{}^{10}C_3} + \frac{{}^4C_3}{{}^{10}C_3} + \dots + \frac{{}^9C_3}{{}^{10}C_3} + \frac{{}^{10}C_3}{{}^{10}C_3} \right]} \\ &= \frac{{}^9C_3}{{}^3C_3 + {}^4C_3 + \dots + {}^9C_3 + {}^{10}C_3} = \frac{{}^9C_3}{{}^{11}C_4} = \frac{14}{55} \end{aligned}$$

$$17.(D) \quad \frac{\sin^{-1}(\cos x) + \cos^{-1}(\sin x)}{\tan^{-1}(\cot x) + \cot^{-1}(\tan x)}$$

$$\begin{aligned} &= \frac{\sin^{-1}\left(\sin\left(\frac{\pi}{2} - x\right)\right) + \cos^{-1}\left(\cos\left(\frac{\pi}{2} - x\right)\right)}{\tan^{-1}\left(\tan\left(\frac{\pi}{2} - x\right)\right) + \cot^{-1}\left(\cot\left(\frac{\pi}{2} - x\right)\right)} = 1 \end{aligned}$$

$$18.(C) \quad L = \lim_{x \rightarrow 0} \frac{f(x^2 + x + 1) - f(1)}{f(x^4 - x^2 + 2x + 4) - f(4)} \quad \left(\frac{0}{0} \text{ form} \right)$$

Since $f(x)$ is differentiable, then applying L' Hospital rule

$$\begin{aligned} L &= \lim_{x \rightarrow 0} \frac{(2x+1)f'(x^2+x+1)}{(4x^3-2x+2)f'(x^4-x^2+2x+4)} \\ &= \frac{f'(1)}{2f'(4)} = 4 \end{aligned}$$

$$19.(C) \quad I = \int \frac{1 + (\sin x)^{2/3}}{1 + (\sin x)^{4/3}} d(\sqrt[3]{\sin x})$$

Putting $(\sin x)^{1/3} = t$, we get

$$I = \int \frac{1+t^2}{1+t^4} dt = \frac{1}{\sqrt{2}} \tan^{-1} \frac{t^2-1}{\sqrt{2}t} + c$$

$$20.(B) \quad I = \int_{1/3}^{2/3} \frac{\log_e x}{\log_e(x-x^2)} dx$$

$$\therefore I = \int_{1/3}^{2/3} \frac{\log_e(x)}{\log_e x + \log_e(1-x)} dx \quad \dots(i)$$

$$\therefore I = \int_{1/3}^{2/3} \frac{\log_e(1-x)}{\log_e(1-x) + \log_e x} dx \quad \dots(ii)$$

$\therefore$ Adding (1) and (2), we get

$$2I = \int_{1/3}^{2/3} 1 \cdot dx \quad \Rightarrow \quad I = \frac{1}{6}$$

SECTION-2

$$21.(9) \quad \sum_{r=0}^{18} {}^{20}C_r \quad {}^{20}C_{r+2} = \sum_{r=0}^{18} {}^{20}C_{20-r} \quad {}^{20}C_{r+2} = {}^{40}C_{22}$$

22.(8) Let 'a' be the first term, 'r' be the common ration and n be the number of terms.

Sum of terms except the first term is:

$$ar \left(\frac{r^{n-1}-1}{r-1} \right) = 63.5 \quad \dots(i)$$

Sum of terms except the last terms is:

$$a \left(\frac{r^{n-1}-1}{r-1} \right) = 127 \quad \dots(ii)$$

Dividing (i) by (ii), we get $r = \frac{1}{2}$

Sum of the terms except the first two terms is:

$$ar^2 \left(\frac{r^{n-2}-1}{r-1} \right) = 31.5 \quad \dots(iii)$$

Dividing (ii) by (iii), we get

$$\frac{1}{r^2} \left(\frac{r^{n-1} - 1}{r^{n-2} - 1} \right) = \frac{127}{31.5}$$

$$\Rightarrow \frac{1}{\left(\frac{1}{2}\right)^2} \left(\frac{\left(\frac{1}{2}\right)^{n-1} - 1}{\left(\frac{1}{2}\right)^{n-2} - 1} \right) = \frac{127}{31.5} \Rightarrow 4 \left(\frac{2\left(\frac{1}{2}\right)^n - 1}{4\left(\frac{1}{2}\right)^n - 1} \right) = \frac{127}{31.5}$$

$$\Rightarrow \frac{2x-1}{4x-1} = \frac{127}{126} \quad (\text{where } (1/2)^n = x)$$

$$\Rightarrow 1 = 256x \Rightarrow x = \left(\frac{1}{2}\right)^8$$

$$\Rightarrow \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^8 \Rightarrow n = 8$$

$$\begin{aligned} 23.(7) \quad & \int e^{\left(\frac{x^6+x^2-x-1}{x-1}\right)} (5x^4 + 4x^3 + 3x^2 + 2x + 2) dx \\ &= \int e^{(x^5+x^4+x^3+x^2+x+1)} \cdot e^x (5x^4 + 4x^3 + 3x^2 + 2x + 1 + 1) dx \\ &= \int e^x (f(x) + f'(x)) dx, \text{ where } f(x) = e^{(x^5+x^4+x^3+x^2+x+1)} \\ &= e^x f(x) + C \\ &= e^x \cdot e^{(x^5+x^4+x^3+x^2+x+1)} + C \end{aligned}$$

$$24.(0) \quad \text{Given } f(x) = \sin^3 x$$

$$\text{Let } p(x) = x^2 + bx + c$$

$$I = \int_0^{2\pi} p(x) f''(x) dx$$

$$= p(x) f'(x) \Big|_0^{2\pi} - \int_0^{2\pi} p'(x) \cdot f'(x) dx$$

$$= 0 - \int_0^{2\pi} p'(x) \cdot f'(x) dx$$

$$= - \left[p'(x) \cdot f(x) \Big|_0^{2\pi} - \int_0^{2\pi} p''(x) \cdot f(x) dx \right]$$

$$= \int_0^{2\pi} p''(x) \cdot f(x) dx$$

$$= 2 \int_0^{2\pi} \sin^3 x dx$$

$$= 0$$

25.(1) Given $f(1) = 2$

$$\alpha^2 + \beta^2 = 1 \text{ and } f(\alpha x) \cdot f(\beta x) = f(x) \forall x$$

Put $\alpha = 1, \beta = 0$ and $x = 1$

$$f(1) \cdot f(0) = f(1) \Rightarrow f(0) = 1$$

Now differentiating $f(\alpha x) f(\beta x) = f(x)$ w.r.t. x and put $x = 0$

$$\alpha f'(0) f(0) + \beta f'(0) f(0) = f'(0)$$

$$(\alpha + \beta) f'(0) = f'(0)$$

$$\Rightarrow f'(0) [\alpha + \beta - 1] = 0 \Rightarrow f'(0) = 0$$

26.(5) Let the number of blue balls be n .

Therefore, the number of black balls $= 10 - (n + 2) = 8 - n$.

If all the three balls drawn are of the same colour, they must be either blue or black, the probability

$$\text{for which is } \frac{{}^nC_3}{{}^{10}C_3} + \frac{{}^{8-n}C_3}{{}^{10}C_3} = \frac{11}{120}$$

$$\Rightarrow n(n-1)(n-2) + (8-n)(7-n)(6-n) = 10 \times 9 \times 8 \cdot \frac{11}{120}$$

$$\Rightarrow 18n^2 - 144n + 336 = 66 \Rightarrow n^2 - 8n + 15 = 0$$

$$\Rightarrow (n-5)(n-3) \Rightarrow n = 5 \text{ (as } n > \text{ number of black balls)}$$

27.(2) Put $x = 1, \omega, \omega^2$ in $(1+x)^{3n}$, where ω is imaginary cube root

$$\text{We get } 3T_n - S_n = \left((-\omega^2)^{3n} + (-\omega)^{3n} \right)$$

$$\Rightarrow |3T_n - S_n| = 2$$

28.(14) α has three choices, B, C and D .

Case I:

If α is going to B , then β has three choices, γ has two choices and δ has only one choice.

The number of ways $= 1 \times 3 \times 2 \times 1 = 6$

Case II:

If α is going to C or D (α cannot go to A), then β has two choice, γ has two choice and δ has only one choice.

The number of ways $= 2 \times 2 \times 2 \times 1 = 8$

Thus, required number ways $= 6 + 8 = 14$.

$$29.(20) \sum_{r=0}^{18} (r^2 + 2r + 1) r!$$

$$= \sum_{r=0}^{18} [(r+2)-1] (r+1)!$$

$$= \sum_{r=0}^{18} [(r+2)! - (r+1)!]$$

$$= 20! - 1$$

30.(8) We have $P^2 = PP = (PQ)P = P(QP) = PQ = P$

Similarly, $Q^2 = QQ = (QP)Q = Q(PQ) = QP = Q$

$$\begin{aligned}\text{Now, } (P+Q)^2 &= P^2 + Q^2 + PQ + QP \\ &= P + Q + P + Q \\ &= 2(P+Q)\end{aligned}$$

$$\Rightarrow |P+Q|^2 = 2^3 |P+Q|$$

$$\Rightarrow |P+Q| = 8 \quad [\text{As det. } (P+Q) \neq 0]$$

$$\begin{aligned}\text{Also, } (P-Q)^2 &= P^2 + Q^2 - PQ - QP \\ &= P + Q - P - Q = 0\end{aligned}$$